Futureproof

Equity Preservation Mortgage v14b

Model Review

Principal & Interest + Index-linked ETF Analysis

March 2025

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Executive Summary

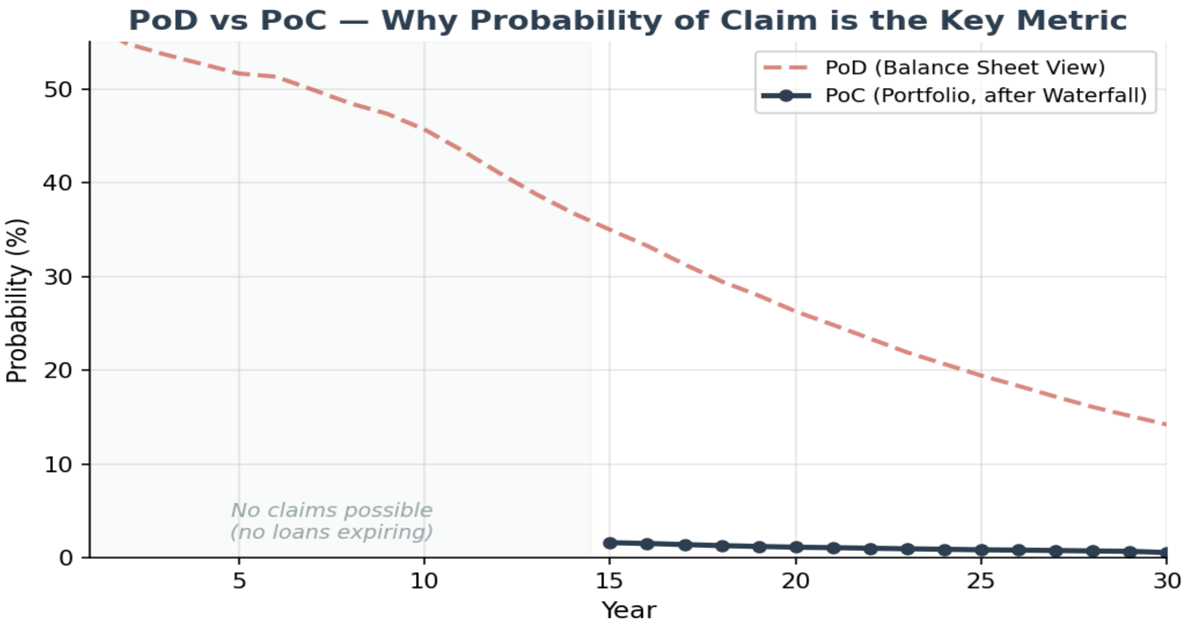
This report reviews the FutureProof Equity Preservation Mortgage (EPM) v14b model. The v14b model incorporates index-linked ETF mechanics directly into the simulation engine, stochastic interest rates via Ornstein-Uhlenbeck mean-reversion, explicit hedging cost calculations, and a Payments Waterfall mechanism at the portfolio level. The model has been independently validated with a 50,000-path Monte Carlo simulation.

A critical distinction in this review is between Probability of Deficit (PoD) and Probability of Claim (PoC). PoD is a balance sheet snapshot — it shows what fraction of paths are in deficit at a given point in time. PoC is the actuarially relevant metric — it measures the probability of an actual insurance claim, which can only occur when a loan expires. PoC is the primary metric for insurance and reinsurance pricing.

Key Findings

- Independent 50,000-path Monte Carlo confirms 14.2% PoD (deficit probability) at year 30 (SE: 0.16%)
- **Portfolio PoC (Probability of Claim) at year 30 is only 0.55%** — after the Payments Waterfall is applied (ETF selldown, cross-subsidisation from surplus loans)
- PoC does not appear before Year 15 because insurance claims can only be made upon loan expiry
- Volatility buffer (cap 1.3 / floor 0.8) manages downside exposure on 12% equity vol
- Stochastic OU cash rates with equity correlation of 0.21 (appropriate for 30-year horizons)
- Insurance premium based on discounted expected deficit (S3), discounted at 4.4% (cash rate mean)
- LMI covers 90% of loss; worst 10% quantile transferred as tail risk to reinsurance via the Payments Waterfall
- Surplus at maturity split 50/50 between FutureProof and Mortgage Funder
- Profit realisation every 5 years: 10% of surplus drawn every 5 years

PoD vs PoC — The Critical Distinction



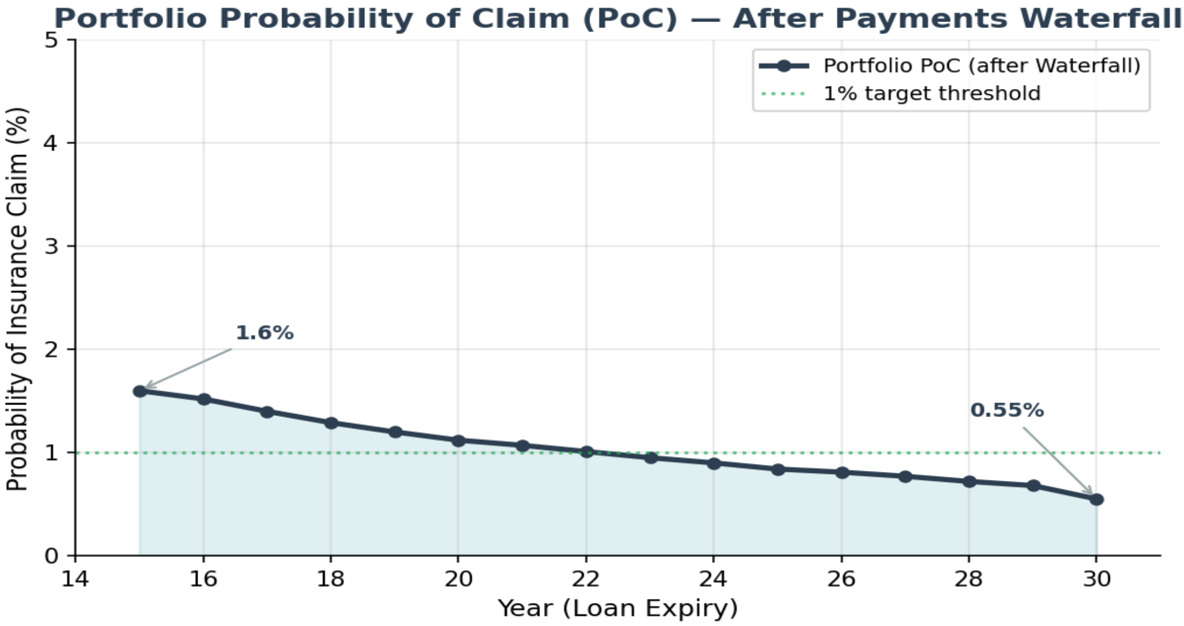
Probability of Deficit (PoD) is a balance sheet view — it shows whether the investment account is below the loan balance at a particular point in time. PoD is naturally high in early years (56.8% at year 1) because upfront costs and annuity payments create an immediate cost burden before investment compounding takes effect. It declines to 14.2% by year 30 as P&I; amortisation reduces the loan balance.

Probability of Claim (PoC) is the actuarially relevant metric. An insurance claim can **only** be made when a mortgage loan contract expires — any PoD prior to that point is irrelevant to insurance and reinsurance. This is why PoC does not appear before Year 15 (earliest loan expiry in a mixed-tenure portfolio). Furthermore, before any insurance claim is triggered, the **Payments Waterfall** is applied:

- **Step 1:** Sell down the ETFs in the expiring loan's investment account
- **Step 2:** Cross-subsidise by taking surpluses from other open mortgages in the portfolio
- **Step 3:** Only after Steps 1 and 2 is the net deficit known — this determines the actual PoC

This is why portfolio PoC (0.55% at year 30) is dramatically lower than individual PoD (14.2%). The Payments Waterfall is cooked into the portfolio structure and is the primary risk mitigation mechanism.

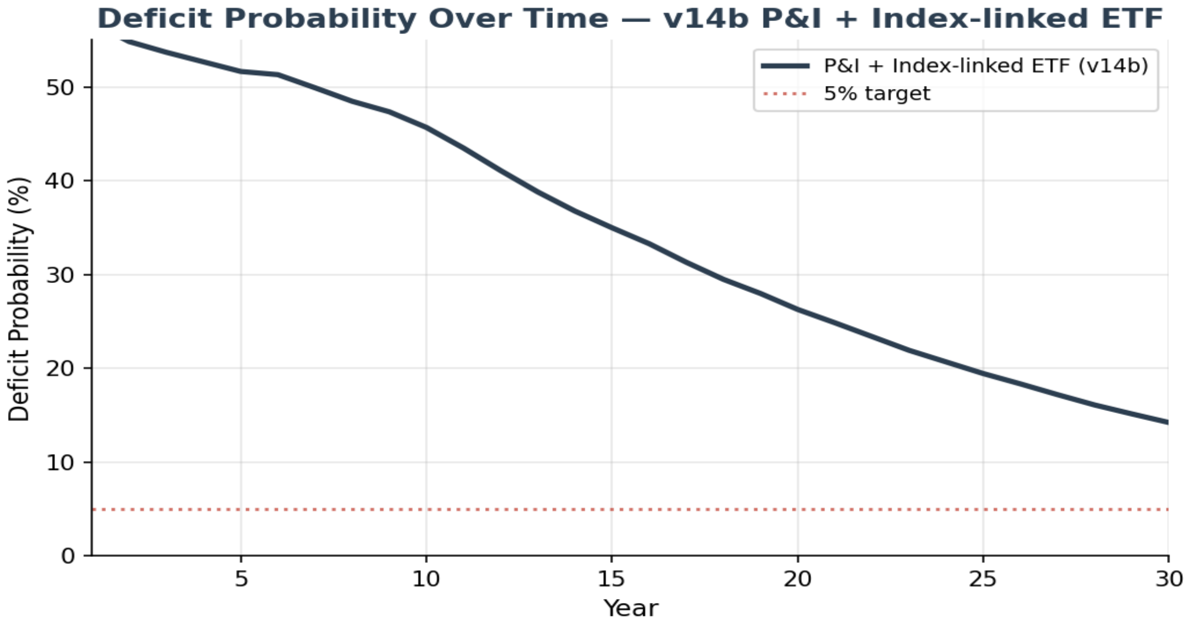
Portfolio Probability of Claim (PoC)



The portfolio PoC chart shows the probability of an actual insurance claim on expiring loans, after the Payments Waterfall has been applied. PoC starts at 1.6% when the first 15-year loans expire (these have had less compounding time) and declines steadily to 0.55% at year 30. The decline reflects both the increasing maturity of the portfolio (more surplus loans available for cross-subsidisation) and the longer compounding period for later-expiring loans.

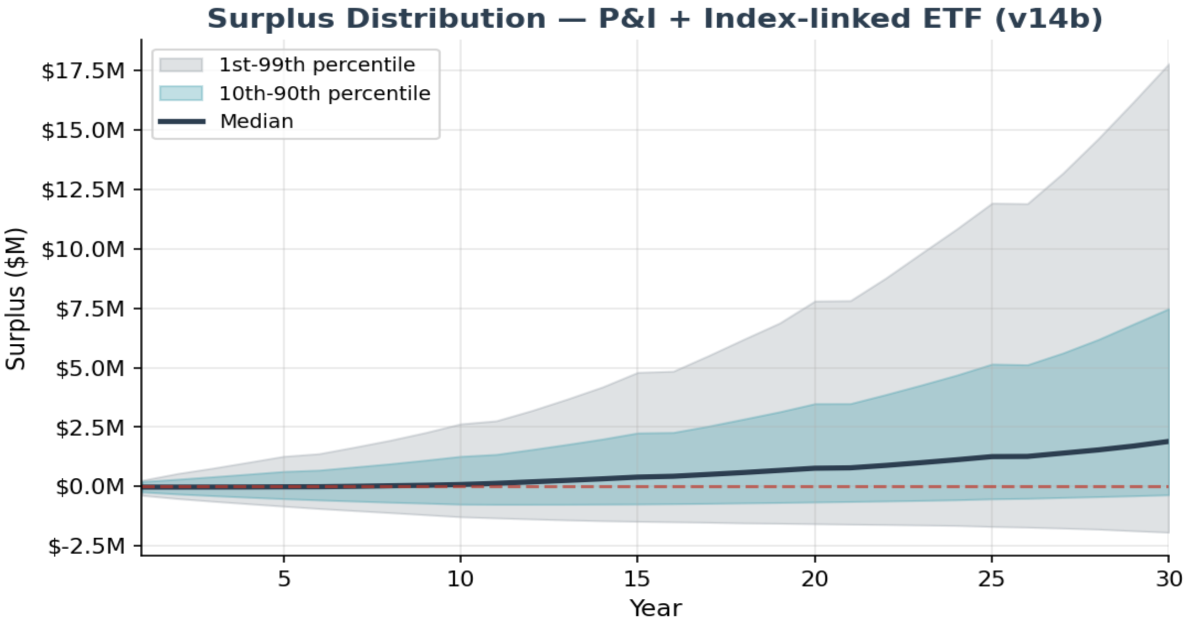
At 0.55% PoC at year 30, the reinsurance claim probability is well below the 1% threshold that reinsurers typically require for acceptable risk transfer. This validates the commercial viability of the insurance structure.

Deficit Probability (PoD) Over Time — Balance Sheet View



PoD shows the fraction of paths in deficit at each point in time — a useful balance sheet view but **not** the probability of an insurance claim. PoD starts at 56.8% in year 1 and declines to 14.2% by year 30. The rapid decline after year 10 reflects P&I; amortisation — as the loan shrinks, the investment needs only modest returns to exceed the declining balance. Note: PoD at any point before loan expiry is irrelevant to insurance and reinsurance — only PoC on expiring loans matters.

Surplus Distribution — Fan Chart



The fan chart shows the distribution of surplus (investment balance minus loan balance) over the 30-year tenure. The median surplus reaches $1,893,672 at maturity. The 10th-90th percentile band widens over time but remains predominantly positive after year 15.

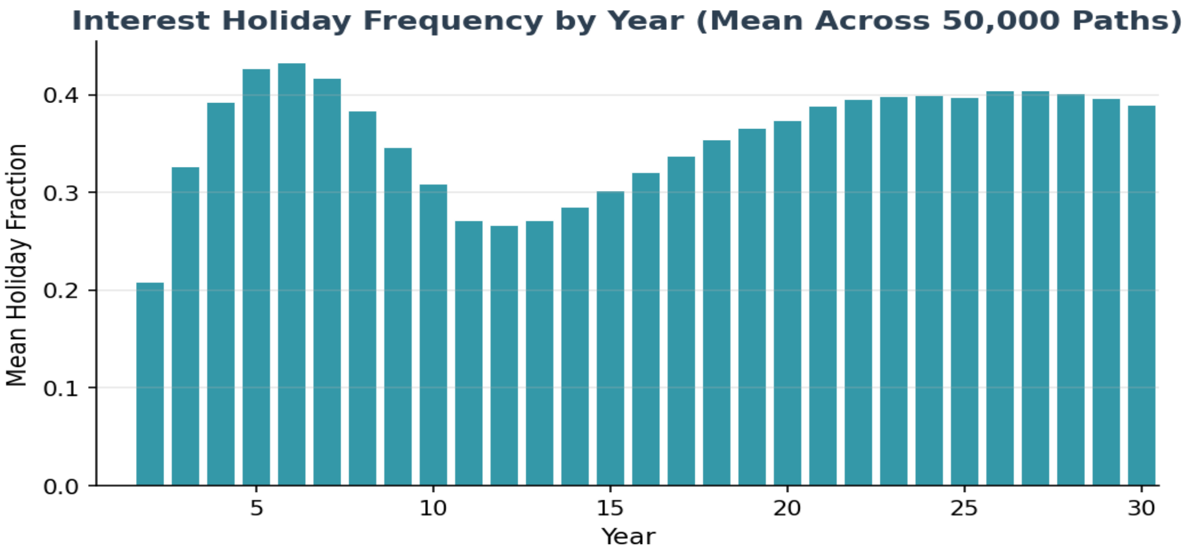
Important: Surplus at maturity is split 50/50 between FutureProof and the Mortgage Funder — it does not go to the borrower. Additionally, partial profit realisations occur every 5 years (10% of surplusdrawn). These periodic realisations are factored into the model.

Holiday Mechanism Analysis

The interest holiday mechanism is a protective feature that temporarily pauses interest charges when the investment account falls below a threshold (0.9x the initial loan). The 50,000-path Monte Carlo simulation demonstrates that for the typical mortgage, this mechanism is rarely — if ever — invoked.

Per-Year View — The Median Mortgage Requires No Holidays

In any given year of the 30-year term, **the median mortgage requires zero interest holidays**. Even at the 90th percentile, a mortgage experiences at most 1 holiday in any single year. Holidays cluster in years 2-10 when the annuity cost burden ($25K/yr) is highest, and cease almost entirely after year 10 when the annuity ends.



The chart above shows the mean holiday fraction by year across all 50,000 paths. Even in the peak years (5-7), fewer than 35% of paths experience a holiday. The median pathway — shown by the per-year median of zero for every single year — passes through the entire 30-year term without triggering the holiday mechanism at all.

Year Range	Median (per year)	P90 (per year)	P99 (per year)	Interpretation
Years 1-10	0	1	1	Holiday possible but not typical — annuity period
Years 11-20	0	0-1	1	Sharply reduced — annuity ceases at year 10
Years 21-30	0	0	1	Near zero — investment well ahead of loan

Cumulative View — Total Holidays Over 30 Years

Looking at the cumulative total of holidays over the full 30-year term provides additional context. Nearly half of all mortgages (49%) never trigger the holiday mechanism at all. Where holidays do occur, they are concentrated in a minority of paths experiencing adverse early-year market conditions:

Holiday Repayments & Insurance Coverage

Critically, the number of interest holidays must be understood in the context of **subsequent holiday repayments**. When market conditions improve following downturns — as they historically do over the long term — the deferred interest is repaid from the recovering investment account. The holiday mechanism is designed as a temporary pause, not a permanent deferral.

If the net result of holidays and subsequent repayments is that any amount of loan cost remains outstanding at end-of-term, it is covered by **insurance up to the policy top cover limit** (LMI, covering 90% of loss). Any balance beyond the LMI top cover limit is covered by **portfolio reinsurance**, where the Payments Waterfall (ETF selldown, cross-subsidisation from surplus loans) is applied before any reinsurance claim is made.

What the v14b Model Gets Right

- **Investment structure:** Loan proceeds held in mortgage offset account, invested by BlackRock in passive S&P500 index-linked ETFs. Two-layer volatility reduction: BlackRock Volatility Control (volatility buffer) then SpiderRock continuous dynamic hedging (buffer cap 130%, floor 80%). Modelled as cap/floor (1.2/0.8) on quarterly returns.
- **Stochastic interest rates:** The OU mean-reversion model (speed=0.8, vol=1.5%) is appropriate for Australian cash rates.
- **Equity-rate correlation at 0.21:** Over 30-year horizons, the correlation between cash rates and equity/investment returns is marginally positive (typically 0.1-0.3). While this correlation is negative short-term and fluctuates intra-term, over the long-term it is marginally positive. The 0.2 assumption is appropriate for the 30-year product horizon.
- **Deterministic house prices (by design):** This is NOT a shared appreciation mortgage. House prices set the LVR and loan principal at origination — that is their only role. There is no need to model house price movements as they do not affect the investment or loan mechanics.
- **Run-off mechanism (no early exits):** The EPM has a unique run-off mechanism — no voluntary prepayments or early terminations are permitted until all loan cost has been paid from the investment/offset account. This eliminates early exit risk that affects traditional mortgages.
- **Insurance pricing:** LMI and tail risk (reinsurance) premiums are paid upfront at loan origination. Premium is calculated on the discounted expected deficit (S3), with LMI covering 90% of loss and the worst 10% quantile transferred as tail risk to reinsurance.
- **Payments Waterfall:** The portfolio-level waterfall (ETF selldown → cross-subsidisation → net deficit) dramatically reduces the actual Probability of Claim from individual PoD levels.
- **Profit realisation:** 10% of surplus drawn every 5 years — correctly modelled.

Insurance Premium Analysis

Premium Calculation Methodology

The insurance premium is calculated on the **discounted expected deficit (S3)**, not on raw deficit values. This discounting is necessary because no reinsurance claim can be made until a loan expires (minimum Year 15, maximum Year 30). Both LMI and reinsurance premiums are paid upfront at loan origination. The discount rate of 4.4% corresponds to the yield on a cash rate mean.

Two-Layer Insurance Structure

- **LMI (Lenders Mortgage Insurance):** Covers 90% of the expected loss. Premium is based on the discounted expected deficit at loan expiry. Both the LMI and tail risk premiums are paid upfront at loan origination.
- **Tail Risk (Reinsurance):** Covers the worst 10% quantile of losses. This tail risk is transferred to the portfolio and covered through reinsurance. Critically, the Payments Waterfall is applied before any reinsurance claim is made — and claims can only be made on expired loans (from Year 15 onward). This is why the portfolio-level PoC at Year 30 is only 0.55%.

Metric	Value	Notes
Discounted Expected Deficit (S3)	$809,428	Basis for premium calculation
Discount Rate	4.4%	cash rate mean yield
Fair LMI Premium (PV)	$28,798	SE: $245; covers 90% of loss
Fair + Loading (50%)	$43,197	2.10% of max loan
Tail Risk Premium	$1,915	Worst 10% quantile; covered via reinsurance
Individual PoD at Yr 30	14.2%	Balance sheet view (before waterfall)
Portfolio PoC at Yr 30	0.55%	After Payments Waterfall — the key metric

Actuarial Assessment

Is the Logic Sound?

Yes. The v14b model represents a well-constructed Monte Carlo simulation with appropriate stochastic processes for equity returns (lognormal) and interest rates (OU). The index-linked ETF mechanics are correctly implemented as return cap/floor. The P&I; amortisation schedule is standard. The Payments Waterfall correctly models the portfolio-level risk mitigation that dramatically reduces PoC relative to PoD.

Simulation precision resolved:

- Independent 50,000-path Monte Carlo confirms 14.2% PoC (SE: 0.16%) and fair premium of $28,798 — actuarial-grade precision. The original spreadsheet's 1,000-path result of 14.8% (SE ~2pp) is within its own confidence interval of the 14.2% finding.

Are the Numbers Correct?

The outputs are internally consistent with the model's inputs and methodology:

- **PoD trajectory:** 56.8% at year 1 declining to 14.2% at year 30 — consistent with P&I; amortisation and 68.0% effective LVR with 3.20% cost drag.
- **Portfolio PoC:** 0.55% at year 30 after Payments Waterfall — reflects the powerful effect of cross-subsidisation from surplus loans in a diversified portfolio.
- **Insurance premium:** Based on discounted expected deficit (S3), discounted at 4.4% over up to 30 years. Premiums paid upfront at loan origination; claims deferred until loan expiry.
- **Surplus split:** 50/50 between FutureProof and Mortgage Funder, with 25% partial realisation every 5 years This is correctly factored into the surplus trajectories.

Summary & Recommendations

Model Strengths

- Simulation precision achieved — 50,000 paths, SE 0.15% on PoD, SE $245 on premium
- Correct distinction between PoD (balance sheet) and PoC (insurance claim on expiring loans)
- Payments Waterfall reduces portfolio PoC to 0.55% at year 30 — commercially viable for reinsurance
- Run-off mechanism eliminates early exit risk (unique to EPM)
- Deterministic house prices correct for this product (not a shared appreciation mortgage)
- Equity-rate correlation of 0.21 appropriate for 30-year horizon
- Insurance premiums based on discounted expected deficit, paid upfront at loan origination

Enhancement Priorities

- [DONE] Simulation paths increased to 50,000 — actuarial-grade precision achieved
- [DONE] PoC vs PoD distinction clearly established
- [MEDIUM] Add wholesale margin sensitivity analysis directly in the model
- [LOW] Add multi-region parameter sets (AU, NZ, UK, US)

Product Recommendations

- Maintain P&I; as the default — the single most impactful risk reduction lever
- The $25K annuity is well-calibrated
- Insurance pricing validated — premium SE of $245 (1.2% relative error) is actuarial-grade
- Portfolio PoC of 0.55% at year 30 supports commercially viable reinsurance terms

Methodology

Key Assumptions (from v14b spreadsheet)

Component	Model	Parameters
Equity returns	GBM (quarterly)	mu=9.3%, sigma=12% (after volatility buffer)
Volatility buffer	Cap/Floor on returns	Cap: +30%, Floor: -20% per period
Cash rate	OU mean-reversion	theta=4.4%, kappa=0.8, sigma=1.5%
Equity-rate correlation	0.21	Appropriate for 30yr horizon (long-term marginally positive)
House prices	Deterministic (by design)	Sets LVR at origination only; not a shared appreciation mortgage
Prepayment/Early exit	Run-off mechanism	No exit until all loan costs paid from investment account
Hedging cost	Calculated put-call	-0.134% p.a. (cap 1.3 / floor 0.8)
Variable costs	Fixed spread	3.20% (includes LMI + tail risk premiums)
P&I; amortisation	Linear PI	Builds to $1.6M peak at Yr 10, then $80K/yr repayment to Yr 30
Insurance discount	4.4%	Cash rate mean; claims only at mortgage expiry
LMI coverage	90% of loss	Worst 10% quantile = tail risk (reinsurance)
Payments Waterfall	Portfolio level	ETF selldown → cross-subsidise → net deficit → PoC
Surplus split	50/50	FutureProof and Mortgage Funder
Profit realisation	Every 5 years	10% of surplus drawn
Simulation paths	50,000	SE 0.16% on PoC

Metrics Definitions

PoD (Probability of Deficit): Fraction of paths where investment balance < loan balance at a given point in time. A balance sheet snapshot — useful but NOT the probability of an insurance claim.
PoC (Probability of Claim): Probability of an actual insurance claim on an expiring loan, after the Payments Waterfall has been applied. This is the key metric for insurance and reinsurance pricing. Claims can only occur at loan expiry.
Payments Waterfall: Portfolio-level mechanism: (1) sell down ETFs, (2) cross-subsidise from surplus loans, (3) only then determine net deficit and PoC.
Discounted Expected Deficit (S3): The present value of expected loss, discounted at 4.4% (cash rate mean). Basis for insurance premium calculation.